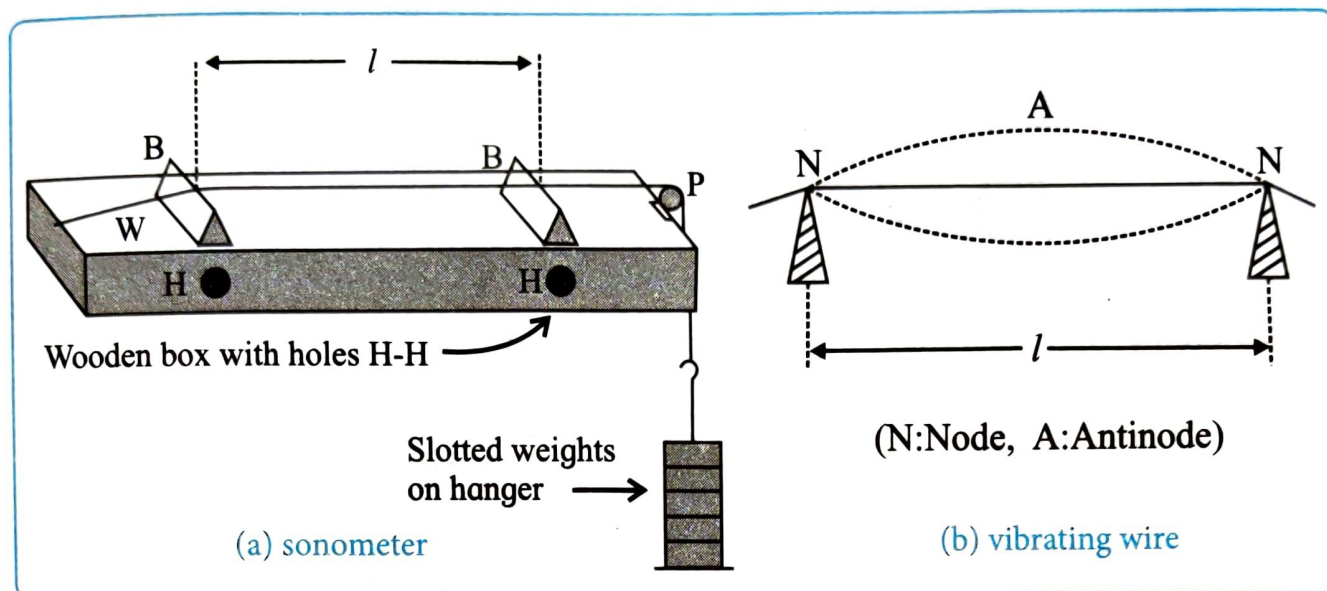


EXPERIMENT NO. 4

SONOMETER - LAW OF LENGTH

Aim: To study the relation between frequency and length of a given wire under constant tension using sonometer.

Apparatus : Sonometer, slotted weights with hanger, set of tuning forks, rubber pad, light paper rider, meter scale.



Theory:

Law of Length: The frequency of stationary transverse vibrations of a stretched string is inversely proportional to its vibrating length when the tension and mass per unit length of the wire are kept constant.

$$\text{Thus } n \propto \frac{1}{\ell} \therefore n\ell = \text{constant when } T \text{ and } m \text{ are constant.}$$

Formula :

$$n = \frac{1}{2\ell} \sqrt{\frac{T}{m}}$$

Where ,

n	=	frequency of vibration of the wire.
T	=	tension applied to the wire.
m	=	mass per unit length of the wire
ℓ	=	vibrating length of the wire.

Procedure :

1. Set up the sonometer on the table and clean the groove on the pulley to ensure that it has minimum friction. Stretch the wire by placing a suitable load on the hanger.
2. Add suitable mass to the hanger to the sonometer wire. Keep it constant throughout the experiment.
3. Place a light paper rider on the wire midway between knife edges.
4. Take a tuning fork of highest frequency, strike it on the rubber pad and set into vibrations.
5. Put the stem of vibrating tuning fork gently on the sonometer box.

6. Slowly, change the distance between two knife edges and adjust particular distance between them so that paper rider falls off.
7. Measure the distance between two knife edges which is the vibrating length of the wire. Take one more reading with the same tuning fork and find mean vibrating length (ℓ)
8. Repeat the procedure with different tuning forks with decreasing order of frequency. Note the corresponding length of vibrating wire.
9. Calculate ' $n\ell$ ' for each tuning fork.
10. Plot a graph of reciprocal of vibrating length $\frac{1}{\ell}$ on (Y-axis) against frequency n on X (axis).
11. Determine unknown frequency from the graph and calculations.
12. Calculate linear density of the wire using mean $n\ell$.

Observations:

Mass of the hanger $M_0 = 500\text{gm}$ Total mass $M = \dots 3500\text{gm}$
Tension applied to the wire $T = \dots 3500 \times 980$ dyne.

To measure vibrating length: =

Obs. No.	Frequency of tuning fork n Hz	Vibrating length of the wire			$\frac{1}{\ell}$	$n\ell$ Hz-cm
		ℓ_1	ℓ_2	Mean ℓ cm		
1	320	31.5	31.2	31.3	$\frac{1}{31.3}$	10016
2	384	29.8	29.6	29.7	$\frac{1}{29.7}$	11404.8
3	426	27.3	27.1	27.2	$\frac{1}{27.2}$	11587.2
4	480	26.8	26.5	26.6	$\frac{1}{26.6}$	12768
5	unkwon $n' = \dots\dots\dots$			$\ell' = \dots\dots\dots$		

Calculations :

To find $n\ell$

Ob. No	1	2	3	4	5
Log $n = a$	2.5051	2.5843	2.6294	2.6812	
Log $\ell = b$	1.4962	1.4728	1.4346	1.4257	
$a + b = c$	4.0013	4.0571	4.0640	4.1069	
Antilog $c = n\ell$	1.003×10^4	1.140×10^4	1.159×10^4	1.279×10^4	

Mean ($n\ell$) = 11444

Umknown frequency $n' = \text{Hz} \frac{(n\ell)_{\text{mean}}}{\ell'} = 332.74 \text{ Hz}$

NOTES / CALCULATIONS / ROUGH WORK

$$1) \frac{31.5 + 31.2}{2} = \frac{62.7}{2} = 31.3$$

$$2) \frac{29.8 + 29.6}{2} = \frac{59.4}{2} = 29.7$$

$$3) \frac{27.3 + 27.1}{2} = \frac{54.4}{2} = 27.2$$

$$4) \frac{26.8 + 26.5}{2} = \frac{53.3}{2} = 26.6$$

$$5) 320 \times 31.35 = 10016$$

$$6) 384 \times 29.7 = 11404.8$$

$$7) 426 \times 27.2 = 11587.2$$

$$8) 480 \times 26.6 = 12768$$

$$\log n = a$$

$$\text{Anti-log } c = nd$$

$$1) 320 = 2.5051$$

$$1) 4.0013 = 1002 + 1 = 1003 \times 10^4$$

$$2) 384 = 2.5843$$

$$2) 4.0571 = 1140 + 0 = 1.140 \times 10^4$$

$$3) 426 = 2.6294$$

$$3) 4.0640 = 1159 = 1.159 \times 10^4$$

$$4) 480 = 2.6812$$

$$4) 4.1069 = 1276 + 3 = 1.279 \times 10^4$$

$$\log d = b$$

$$1) 31.35 = 4.962$$

$$2) 29.7 = 4.728$$

$$3) 27.2 = 4.346$$

$$4) 26.6 = 4.257$$

$$A + b = c$$

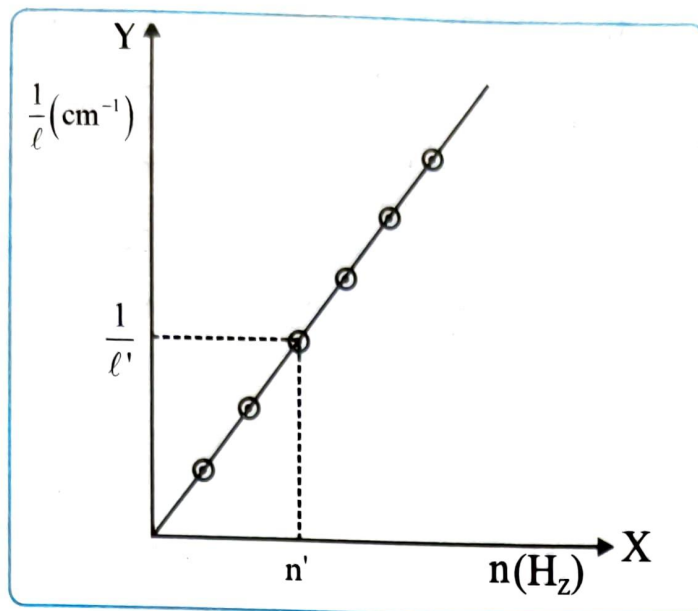
$$1) 2.5051 + 1.4962 = 4.0013$$

$$2) 2.5843 + 1.4728 = 4.0571$$

$$3) 2.6294 + 1.4346 = 4.0640$$

$$4) 2.6812 + 1.4257 = 4.1069$$

Graph : A graph of $1/\ell$ against n



Scale Y = -----cm⁻¹
X = -----Hz

Result :

1. $n\ell = \dots 1.4.4.4 \dots = \text{constant}$, when T and m are kept constant, Hence law of length is verified.
2. The graph of reciprocal of vibrating length against frequency of vibrations of wire is straight line. Hence $n \propto \frac{1}{\ell}$
3. Unknown frequency of tuning fork by calculation $n' = \dots \dots \dots$ Hz
4. Unknown frequency of tuning fork by graph $n' = \dots \dots \dots$ Hz

Precautions :

1. Place the paper rider mid – way between knife edges.
2. Paper rider should be light and dry.
3. Strike tuning fork on the rubber pad gently.
4. Place the stem of the vibrating tuning fork gently on the sonometer box.
5. Measure the distances between the knife-edges accurately.

Questions

1. State the law of length of vibrating string.

The fundamental frequency of a stretched string is inversely proportional to its vibrating length, provided the tension and mass per unit length remain constant.

$FL = \text{constant}$

2. What is fundamental mode of vibration?

The mode of vibration in which the string vibrates as a single segment with nodes at the two ends and one antinode at the centre is called the fundamental mode of vibration (or first harmonic). It has the lowest frequency of vibration.

3. What is resonance?

Resonance is the phenomenon in which the amplitude of forced vibrations becomes maximum when the frequency of the external periodic force is equal to the natural frequency of the vibrating body.

4. What is forced vibrations?

The vibrations of a body produced and maintained by an external periodic force are called forced vibrations.

5. Explain why holes are kept on one side of sonometer box?

Holes are provided on one side of the sonometer box so that the air inside the box can vibrate along with the string. This increases the intensity of sound due to resonance and makes the sound louder.

Remark and sign of teacher :